

A Domain-Specific Language for Customizable Comparison Logic in FactCheck

Create, validate, and execute expert-defined comparison logic.

Praktikum Informatik 2

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The Problem: Different Sources Describe the Same Data Differently

Attribute	Source A	Source B
Title	RoboCop	RoboCop (Blu-ray)
Release Date	1987-07-17	July 17, 1987
Duration	102 min	118 min

**Do these records
describe the same entity?**

The Problem: Simple Equality Is Not Enough

Attribute	Source A	Source B	Exactly Equal?
Title	RoboCop	RoboCop (Blu-ray)	✗
Release Date	1987-07-17	July 17, 1987	✗
Duration	102 min	118 min	✗

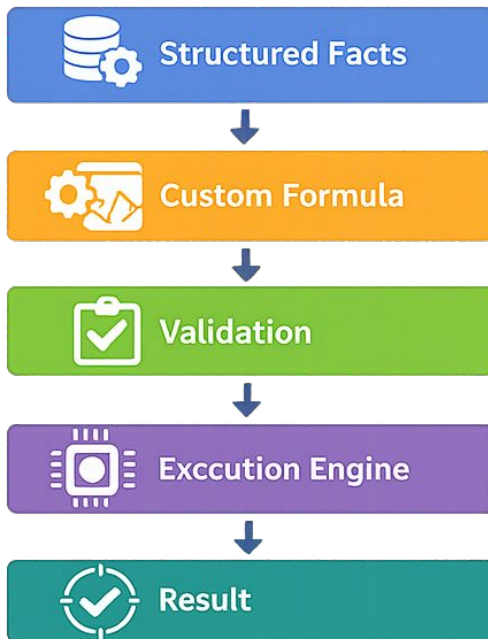
What should be the final decision?

- MATCH?
- SIMILAR?
- MISMATCH?

Solution: Customizable Fact Comparison Logic

Experts define comparison formula using predefined functions.

Attribute	Source A	Source B	Exactly Equal?
Title	RoboCop	RoboCop (Blu-ray)	✗
Release Date	1987-07-17	July 17, 1987	✗
Duration	102 min	118 min	✗



Movie Metadata Consistency Check formula:

```
normalizedTitleMatch(name1, name2)
AND
sameYear(datePublished1, datePublished2)
AND
rangeAbsolute(duration1, duration2, 20)
```

Example Comparison Result

Execution Result

Movie Metadata Consistency Check

version 1

24/06/2026, 08:11:07

Result: **TRUE**

Label: **SIMILAR**

Explanation

Title normalization

"RoboCop" -> "robocop"

"Robocop (Blu-ray)" -> "robocop"

Normalized titles are equal.

Publication year match

1987-07-17 -> 1987

July 17, 1987 -> 1987

Same year **TRUE**

Absolute range comparison

Base value 102

Compared value 118

Difference 16

Allowed difference 20

Within allowed range **TRUE**

Confidence exact match =
100%, tolerance
boundary = 50%,
calculated = 60%

Confidence

87%

Calculated from the evidence scores in the
explanation.

Execution Time

1.014 ms

Technology & Technical Challenge

Technology

Frontend



Backend



Database



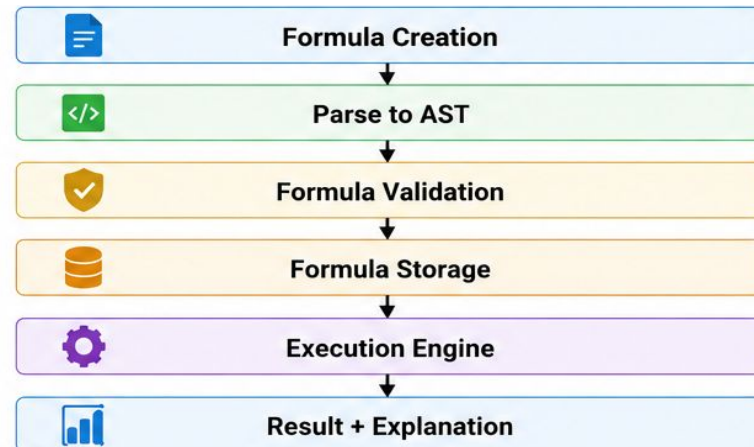
Containerization



Main Technical Challenge

Safely validating and executing user-defined formulas.

User-defined formulas are parsed, validated, stored, and executed through a controlled execution pipeline.



Visit my table to see customizable comparison logic in action.